World Gate Maker 1.2.1 - User Manual

Installing World Gate Maker - read this first

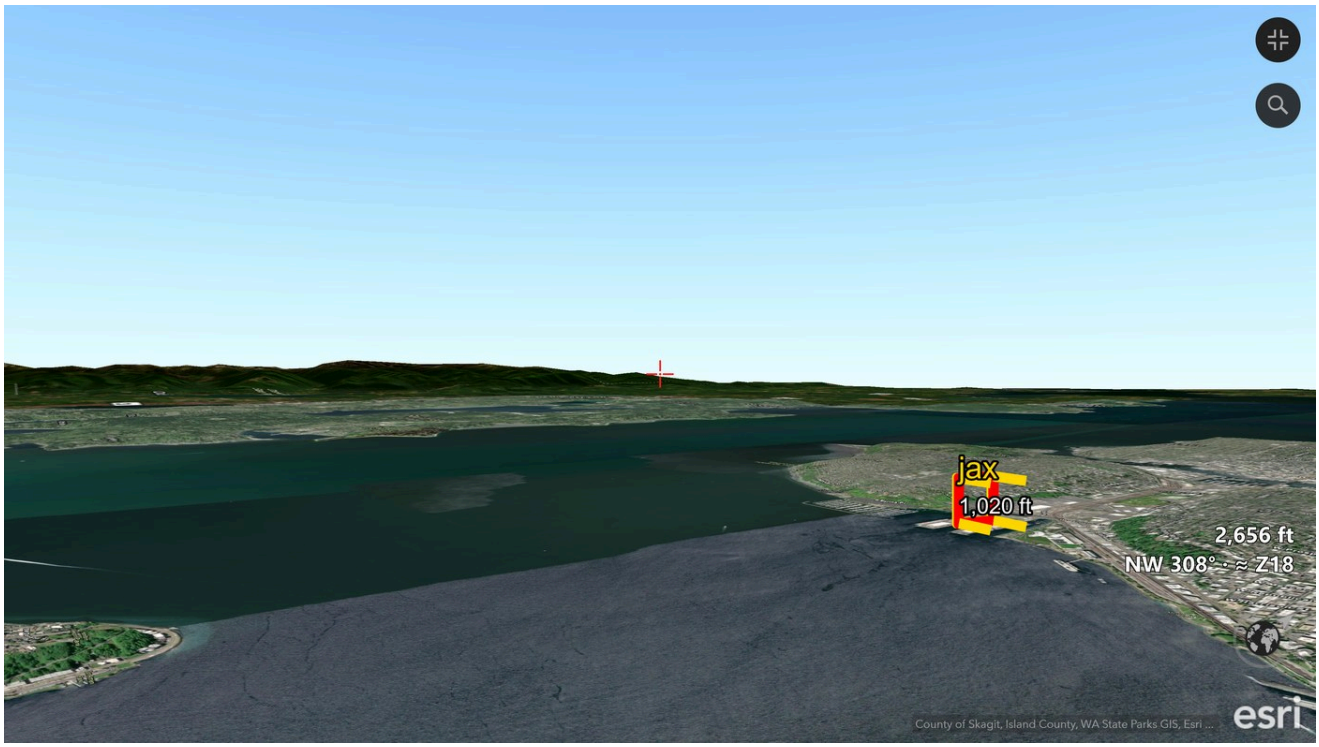
1. **Extract All first:** right-click the downloaded zip > **Extract All**, then open the new folder. (Double-clicking files while they are still inside the zip does not work.)
2. **Double-click** `World_Gate_Maker.bat` - the only .bat in the folder. This manual sits right next to it. The **System Files** folder holds the program - leave it alone.
3. **Blue "Windows protected your PC" box?** Click **More info**, then **Run anyway**. It only asks the first time. (Or, before extracting: right-click the zip > **Properties** > tick **Unblock** > **OK**.)
4. **Python - nothing to install by hand.** If Python is missing, the window asks **Install Python now? [Y/N]**: type **Y** and press Enter. It installs just for you (no admin needed). On an older PC this can take several minutes and Windows Defender may make the PC look frozen - **do not close the window**.
5. Extract it next to your EarthStick folder (both in Downloads, say) and it finds EarthStick's **Gates** folder by itself.
6. To fly through the gates you need EarthStick; to see them, **ArcGIS Earth** (free from Esri: https://www.esri.com/en-us/arcgis/products/arcgis-earth/downloads).

Something stops it? The window says what to do in plain words, and keeps a log in `System Files\Ready_To_Fly_log.txt`. On a work computer that blocks installs, ask IT to install Python 3 from python.org.

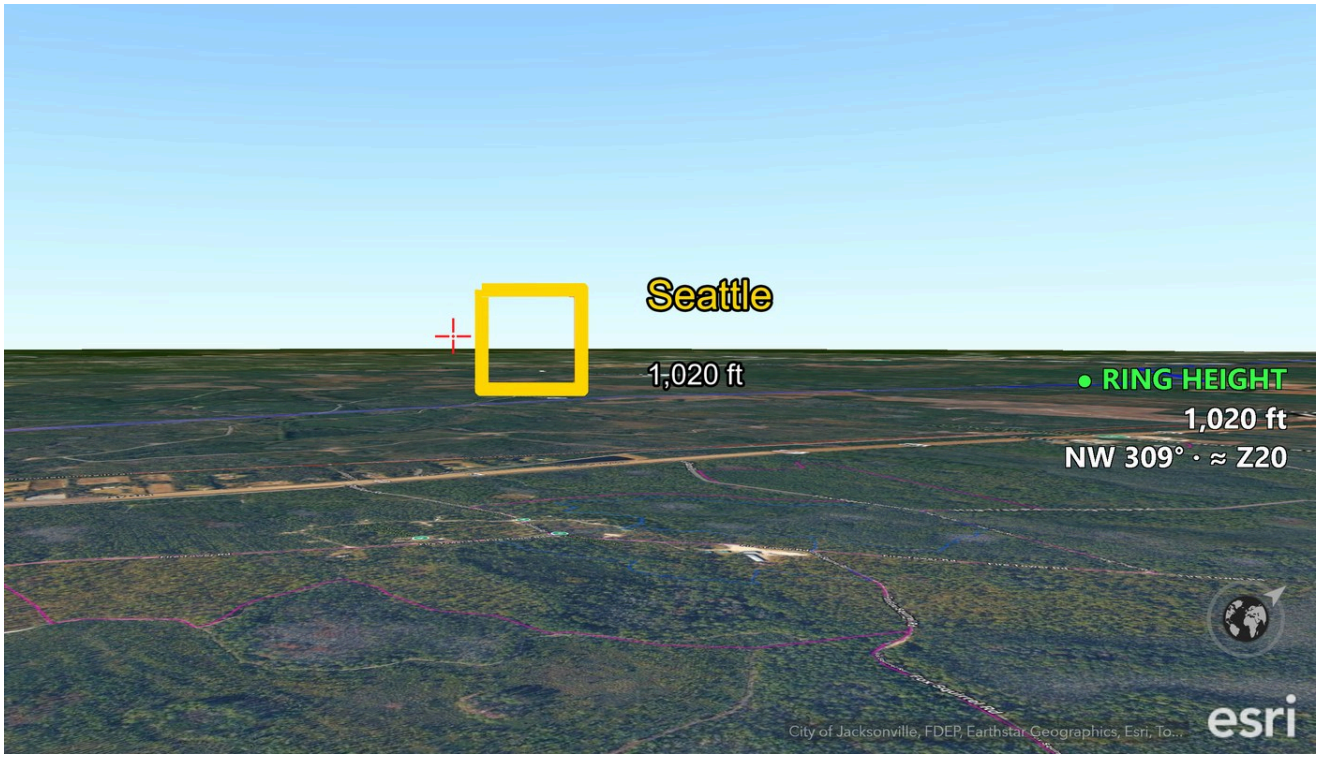
Printable copy: World_Gate_Maker_User_Manual.pdf

Make your own **World Gates**: pick two places, and World Gate Maker builds a pair of golden gates. Fly through the gate at one place with EarthStick and you come out at the other; the gate there brings you back.

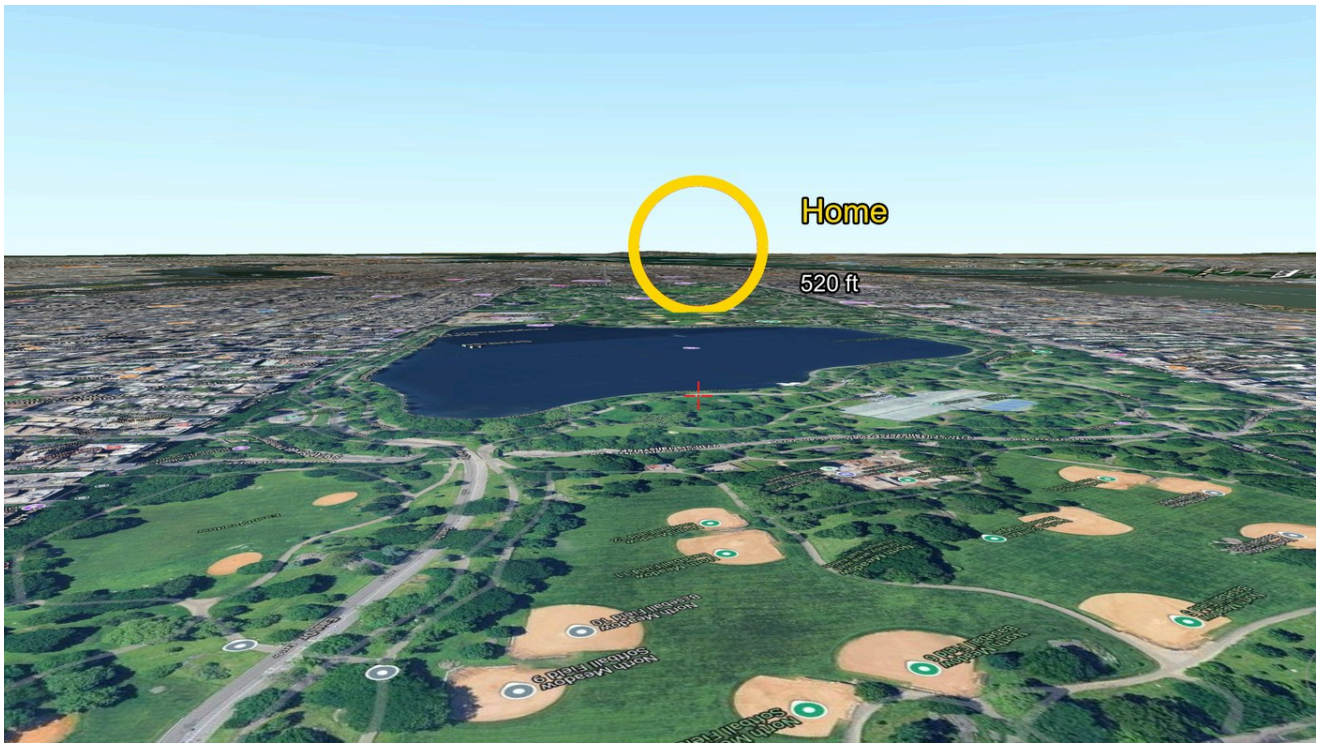
New in 1.1: the gates are **squares with four legs** running back from the corners - the same look as the World Gates pack in EarthStick 1.3.4. One glance shows where the way in is and which way the gate points.



Jacksonville to Seattle with World Gate Maker 1.1: just arrived in Seattle, with the "jax" gate home seen from the side - the legs show the way in at a glance.



Safe return home: lined up on the "Seattle" gate in Jacksonville, at the gate's height (1,020 ft) - EarthStick 1.3.4's green RING HEIGHT light is on.



Made with World Gate Maker 1.0 (when gates were rings): a "Home" gate over Central Park, partnered with a "Central Park" gate near home in Jacksonville.

What you need

- **EarthStick 1.3 or later** with its **Gates** folder (EarthStick page).
- **ArcGIS Earth** desktop with the Automation API on (as for EarthStick).
- **Python 3** - since 1.2.1 `World_Gate_Maker.bat` installs it for you if it is missing (answer **Y**; no admin needed). World Gate Maker uses only the standard library: nothing else to install.

Start

1. Download `World_Gate_Maker_v1.2.1.zip`, right-click it > **Extract All**.
2. Open the `World_Gate_Maker_v1.2.1` folder and double-click `World_Gate_Maker.bat`. (Beside it: this manual as a PDF. `System Files` holds the program - leave it alone.)

Tip: extract it next to your EarthStick folder (for example both in Downloads) and World Gate Maker finds EarthStick's **Gates** folder by itself.

What's in the zip



Make a pair of gates

For **Place 1** and **Place 2**:

1. **Name** - what the place is called, for example `Home`, `Station 7`, `Central Park`. Keep names short (18 characters or fewer) so ArcGIS Earth shows them on one line.
2. **Latitude / Longitude** - type them, or use Google Maps: right-click the spot, click the numbers at the top of the menu to copy them, then press **From Clipboard**. The line underneath shows how they were read - check it.
3. **Ground elev (ft)** - in ArcGIS Earth, put the mouse pointer on the spot and read **elev** in the status bar at the bottom. Type that number.

Then:

4. **Height above ground (ft)** - how high the middle of each gate floats above the ground (default 500 ft).
5. **Gate size** - the width of the square: Small (70 m), Medium (180 m, the default) or Large (320 m, the size of the World Gates pack). The legs grow with it: 71 m, 178 m and 320 m.
6. Press **MAKE GATES**. The first time, choose the **Gates** folder inside your EarthStick folder; World Gate Maker remembers it.

It saves `Gate_<Place 1>_<Place 2>.kmz` and shows the eye alt to fly at for each gate.

Fly them

1. Drag the new `.kmz` from the Gates folder onto ArcGIS Earth so you can see the gates.
2. **Restart EarthStick** - it reads the Gates folder when it starts. The console lists the new gates.
3. Fly to a gate, line up on its **gold** side, match the **eye alt** shown under its name, and fly through the middle. Lined up straight, the four legs make an even X; if they lean, you are off to that side or too high or low.
4. With **EarthStick 1.3.4 or later**, a green **RING HEIGHT** light comes on when you are at the gate's height, and on the last 1,500 m the tractor beam pulls you onto it (**LOCKED ON**). The beam grabs you when you are within the gate's grab zone above or below its middle: Small ±164 ft, Medium ±410 ft, Large ±738 ft. Higher or lower than that, it lets you fly past. Hold Button 3 or 4 to break free.

How the gates are set up

- Each gate is **named for where it takes you**: the gate at Place 1 carries Place 2's name, and the gate at Place 2 carries Place 1's name.
- Each gate **faces the other place**, so you fly toward home to go home.
- **Gold front, red back** - from the red side nothing happens. The legs point out the back (the way you leave).
- You arrive **300 m in front of the other gate**, heading away from it, so you never bounce straight back.
- Names and eye alt numbers show only when you are near (within about 4 miles), so far-away views stay clean.

Messages

- **"The two places need different names"** - each gate is named for its destination, so the names must differ.
- **"Height above ground must be at least ... ft"** - the bottom of the gate would touch the ground. Raise the height or pick a smaller gate.
- **"The places are only ... m apart"** - the places are too close for that gate size.
- **"... is long"** - a name over 18 characters may split onto two lines in ArcGIS Earth. Shorten it if it does.
- **"... is not named 'Gates'"** - EarthStick only reads the Gates folder beside `EarthStick.py`. Choose that folder.

Turning gates off

Move the `.kmz` out of the Gates folder (and remove it from ArcGIS Earth), then restart EarthStick.

Self-test

From a command prompt in the World Gate Maker folder's **System Files**:

```
py -3 World_Gate_Maker.py --selftest
```

Checks coordinate reading, gate names and directions, heights, the safety limits, the square and its legs, and the saved file.